

RAHAMATHULLA REEMAZ H

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 Coimbatore, Tamil Nadu



 [Rahamathulla Reemaz H](#)

Entry-level **Electronics and Communication Engineering** graduate with a keen interest in Firmware Development and Embedded Systems, eager to learn, grow, and contribute in a professional engineering environment.

EDUCATION

B.E ELECTRONICS AND COMMUNICATION ENGINEERING, <i>Karpagam College of Engineering</i> CGPA – 7.52	2023–2027 Coimbatore, India
ST. JOSEPH’s MATRIC.HR.SEC SCHOOL	
HIGHER SECONDARY EDUCATION, MATRIC – 88.7%	2023 Nagercoil, India
SECONDARY EDUCATION, MATRIC-PASS	2021 Nagercoil, India

SKILLS

HARDWARE	PROGRAMMING	CORE TECHNICAL
8051 Microcontroller - Arduino Uno - ESP32	- C Programming - Data structures	- Electrical circuit - Digital electronics - Embedded Systems
SOFTWARE USED	PROTOCOLS	SOFT SKILLS
- Arduino IDE - STM32 Cube IDE	- UART - SPI - I2C	- Team Collaboration - Critical Thinking - Adaptability - Effective Communication

CERTIFICATION

- Altium Education PCB basic design course - CISCO - Introduction to Internet of Things - CISCO – Network Essentials - CISCO – C Essentials
NPTEL: - Digital Electronic Circuits - Introduction to Industry 4.0 and Industrial Internet of Things – Elite
Communication - Speaking test in Linguaskill: CEFR Level -B2 - Writing test in Linguaskill: CEFR Level-B1 - TCS iON –Email Etiquette
Internships Intern Certify: Domain- Java Programming

PROJECTS

Smart Rain and Heat Response Window and Cooler System

- Designed an automated system using rain and temperature sensors to monitor environmental conditions in real time.
- Automatically closed windows during rainfall and activated cooling during high temperatures to improve comfort and energy efficiency.

Smart Customer Support Chatbot Using Machine Learning and GPT

- Developed an AI-powered chatbot using machine learning and GPT to handle customer queries intelligently.
- Improved response accuracy and reduced support response time through context-aware conversations.

Reverse Parking Assist System Using Arduino IDE

- Implemented a reverse parking system using ultrasonic sensors to measure obstacle distance in real time.
- Provided audio and visual alerts to enhance driver safety during parking.

Intelligent Adaptive Real-Time ECG Monitoring with Medical-Aware Scheduling

- Developed an Intelligent Adaptive Real-Time ECG Monitoring System using Raspberry Pi and biomedical sensors (AD8232, MAX30102, DS18B20) to monitor ECG, heart rate, SpO₂, and body temperature in real time.
- Implemented multi-threaded real-time scheduling using POSIX threads and SCHED_FIFO policy to ensure continuous ECG signal acquisition and reliable processing under dynamic workloads.
- Designed a predictive CPU utilization control algorithm that dynamically adjusts task priorities and execution periods to prevent system overload and reduce deadline misses.
- Integrated AI-based health analytics and medical-aware scheduling to detect abnormal cardiac patterns and automatically prioritize life-critical ECG processing with alert mechanisms.

DECLARATION

I hereby declare that the information provided in this resume is true to the best of my knowledge.